

## PROBLEMS ON SIMPLEX METHOD

Q) solve the following LPP by SIMPLEX METHOD

minimize  $Z = x_1 - 3x_2 + 3x_3$ , subject to

$$3x_1 - x_2 + 2x_3 \leq 7,$$

$$2x_1 + 4x_2 \geq -12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

$$x_1, x_2, x_3 \geq 0$$

convert the given problem to Standard form

condition ①: Objective function must be of MAXIMIZATION type

condition ②: All constraints should be expressed as EQUATIONS

condition ③: RHS of each constraint has to be non-negative

condition ④: All the variables  $(x_1, x_2, \dots, x_n)$  must be non-negative

Soln:-

condition-①

Page-②

Given problem is MINIMIZING the objective function

$$Z = x_1 - 3x_2 + 3x_3.$$

converting it to maximization form, we set the objective function as

$$\max Z' = \min(Z)$$

↓

$$Z' = -Z$$

$$\therefore \max Z' = -x_1 + 3x_2 - 3x_3$$

$\min Z$   
is equivalent to  
 $\max (-Z)$

Let us check condition - (3) first before (2),

page - (3)

condition - (3)

RHS of each constraint must be non-negative

From the second constraint, we have

$$2x_1 + 4x_2 \geq -12$$

-ve

multiply by (-1),

$$-2x_1 - 4x_2 \leq 12$$

condition-(2)

All the constraints must be expressed as EQUATIONS

First constraint:

$$3x_1 - x_2 + 2x_3 \leq 17$$

This is equivalent to

$$3x_1 - x_2 + 2x_3 + s_1 = 17$$

Here  $s_1 \geq 0$  is known as  
SLACK VARIABLE

second constraint:

$$-2x_1 - 4x_2 \leq 12$$

This is equivalent to

$$-2x_1 - 4x_2 + s_2 = 12$$

↓  
 $s_2 \geq 0$  is slack variable

Third constraint:

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

This is equivalent to

$$-4x_1 + 3x_2 + 8x_3 + s_3 = 10$$

↓  
 $s_3 \geq 0$  is slack variable



condition-④

All the variables must be non-negative.

This condition is already satisfied.

∴ given that  $x_1 \geq 0$ ,  $x_2 \geq 0$  &  $x_3 \geq 0$

∴ Given linear programming problem is

$$\max Z' = -x_1 + 3x_2 - 3x_3 + 0s_1 + 0s_2 + 0s_3$$

subject to:

$$3x_1 - x_2 + 2x_3 + s_1 + 0s_2 + 0s_3 = 7 \quad \text{---①}$$

$$-2x_1 - 4x_2 + 0x_3 + 0s_1 + s_2 + 0s_3 = 12 \quad \text{---②}$$

$$-4x_1 + 3x_2 + 8x_3 + 0s_1 + 0s_2 + s_3 = 10 \quad \text{---③}$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

Initially set  $x_1 = x_2 = x_3 = 0$  (non-basic solution) Page-(7)

When  $x_1 = x_2 = x_3 = 0$ , we get

From Eqn ①:  $s_1 = 7$

From Eqn ②:  $s_3 = 10$

From Eqn ③:  $s_2 = 12$

BASIC SOLUTION

consider the following table:

CHECK THE OBJECTIVE FUNCTION  
 $z' = -x_1 + 3x_2 - 3x_3 + 0s_1 + 0s_2 + 0s_3$

| $C_B$                   | $C_j$<br>basis | -1<br>$x_1$ | 3<br>$x_2$ | -3<br>$x_3$ | 0<br>$s_1$ | 0<br>$s_2$ | 0<br>$s_3$ | $b$ | $\theta$ |
|-------------------------|----------------|-------------|------------|-------------|------------|------------|------------|-----|----------|
| 0                       | $s_1$          |             |            |             |            |            |            |     |          |
| 0                       | $s_2$          |             |            |             |            |            |            |     |          |
| 0                       | $s_3$          |             |            |             |            |            |            |     |          |
| $Z_j = \sum C_B a_{ij}$ |                |             |            |             |            |            |            |     |          |
| $G_j = C_j - Z_j$       |                |             |            |             |            |            |            |     |          |

From constraints ①, ② & ③, we have

page-(8)

$$A = \begin{bmatrix} 3 & -1 & 2 & 1 & 0 & 0 \\ -2 & -4 & 0 & 0 & 1 & 0 \\ -4 & 3 & 8 & 0 & 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 7 \\ 12 \\ 10 \end{bmatrix}$$

incorporate these rows in the table

| $C_B$                   | $C_j$<br>BASIS | -1<br>$x_1$ | 3<br>$x_2$ | -3<br>$x_3$ | 0<br>$s_1$ | 0<br>$s_2$ | 0<br>$s_3$ | b  | $\theta$ |
|-------------------------|----------------|-------------|------------|-------------|------------|------------|------------|----|----------|
| 0                       | $s_1$          | 3           | -1         | 2           | 1          | 0          | 0          | 7  |          |
| 0                       | $s_2$          | -2          | -4         | 0           | 0          | 1          | 0          | 12 |          |
| 0                       | $s_3$          | -4          | 3          | 8           | 0          | 0          | 1          | 10 |          |
| $Z_j = \sum C_B a_{ij}$ |                |             |            |             |            |            |            |    |          |
| $C_j = c_j - Z_j$       |                |             |            |             |            |            |            |    |          |



NOW, construct the row  $Z_j = \sum C_B a_{ij}$

Example  
1st element:  $(0)(3) + 0(-2) + 0(-4) = 0$

$\downarrow$   
 $C_B$        $\downarrow$   
            Find  
            column

NEXT, construct the row  $C_j = C_j - Z_j$

$\downarrow$   
TOP

| $C_B$                   | $C_j$<br>BASIS | -1<br>$x_1$ | 3<br>$x_2$ | -3<br>$x_3$ | 0<br>$s_1$ | 0<br>$s_2$ | 0<br>$s_3$ | b  | $\theta$ |
|-------------------------|----------------|-------------|------------|-------------|------------|------------|------------|----|----------|
| 0                       | $s_1$          | 3           | -1         | 2           | 1          | 0          | 0          | 7  |          |
| 0                       | $s_2$          | -2          | -4         | 0           | 0          | 1          | 0          | 12 |          |
| 0                       | $s_3$          | -4          | 3          | 8           | 0          | 0          | 1          | 10 |          |
| $Z_j = \sum C_B a_{ij}$ |                | 0           | 0          | 0           | 0          | 0          | 0          | 0  |          |
| $C_j = C_j - Z_j$       |                | -1          | 3          | -3          | 0          | 0          | 0          |    |          |

$\uparrow$  LARGEST +ve  $C_j$  value

Observe that the second column (column corresponding to  $x_2$ ) has the largest free value of  $G_j = c_j - z_j$  page-(10)

∴ This column is considered as KEY COLUMN &  $x_2$  is known as INCOMING VARIABLE

Now, we have to find the OUTGOING VARIABLE by considering the last column  $\theta$ .  
The ' $\theta$ ' column can be obtained by dividing the  $\theta$  column entries from the corresponding elements of KEY COLUMN.

cost of  $x_2$  is 3

| $C_B$                   | $C_j$<br>BASIS | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $b$ | $\theta$                 |
|-------------------------|----------------|-------|-------|-------|-------|-------|-------|-----|--------------------------|
| 0                       | $s_1$          | 3     | -1    | 2     | 1     | 0     | 0     | 7   | $7/-1 = -7$              |
| 0                       | $s_2$          | -2    | -4    | 0     | 0     | 1     | 0     | 12  | $12/-4 = -3$             |
| 0                       | $s_3$          | -4    | 3     | 8     | 0     | 0     | 1     | 10  | $10/3 = 3.33 \leftarrow$ |
| $Z_j = \sum C_B a_{ij}$ |                | 0     | 0     | 0     | 0     | 0     | 0     | 0   |                          |
| $C_j = C_j - Z_j$       |                | -1    | 3     | -3    | 0     | 0     | 0     |     |                          |

KEY ELEMENT

minimum  
POSITIVE  
RATIO

∴ outgoing  
variable is

$s_3$

In next table, we drop  $s_3$  &  
introduce  $x_2$  with its associated  
value 3 under  $C_B$  column.

PAGE (19)

Now, convert the key element to unity using the operation  $R_3 \rightarrow \frac{R_3}{3}$

$$R_3 \rightarrow \frac{R_3}{3}$$

[illegible]

## OPERATIONS

$$R_2 \rightarrow R_2 + 4 \text{ new } R_3$$

[illegible]



NOU compute  $Z_j = \sum C_B a_{ij}$  &  $C_j = C_j - Z_j$  PAG (14)

| $C_B$                   | $C_j$<br>BASIS | -1<br>$x_1$ | 3<br>$x_2$ | -3<br>$x_3$ | 0<br>$s_1$ | 0<br>$s_2$ | 0<br>$s_3$ | $b$    | $\theta$ |
|-------------------------|----------------|-------------|------------|-------------|------------|------------|------------|--------|----------|
| 0                       | $s_1$          | $5/3$       | 0          | $14/3$      | 1          | 0          | $1/3$      | $31/3$ |          |
| 0                       | $s_2$          | $-22/3$     | 0          | $32/3$      | 0          | 1          | $4/3$      | $76/3$ |          |
| 3                       | $x_2$          | $-4/3$      | 1          | $8/3$       | 0          | 0          | $1/3$      | $10/3$ |          |
| $Z_j = \sum C_B a_{ij}$ |                | -4          | 3          | 8           | 0          | 0          | 1          | 10     |          |
| $C_j = C_j - Z_j$       |                | 3           | 0          | -11         | 0          | 0          | -1         |        |          |

↑ LARGEST the value  $\therefore x_1$  is the INCOMING VARIABLE

The first element of  $Z_j$  is computed as follow

$$0 \cdot \left(\frac{5}{3}\right) + 0 \cdot \left(\frac{-22}{3}\right) + 3 \cdot \left(\frac{-4}{3}\right)$$

observe  $C_B$  column  
1st column

second element:

$$0 \cdot (0) + 0 \cdot (0) + 3 \cdot (1) = 3$$

$C_B$  column  
second column

Observe that the first column (column corresponding to  $x_1$ ) has the largest possible value of  $c_j - z_j$ . This column is considered as KEY COLUMN &  $x_1$  is known as INCOMING VARIABLE

Now, we have to find the OUTGOING VARIABLE by considering the last column  $\theta$ .

The  $\theta$  column can be obtained by DIVIDING the  $b$  column entries from the corresponding elements of KEY COLUMN

| $C_B$       | $C_j$<br>BASIS | -1<br>$x_1$                                                                  | 3<br>$x_2$ | -2<br>$x_3$ | 0<br>$s_1$ | 0<br>$s_2$ | 0<br>$s_3$ | $b$    | $\theta$                                 |
|-------------|----------------|------------------------------------------------------------------------------|------------|-------------|------------|------------|------------|--------|------------------------------------------|
| 0           | $s_1$          | <span style="border: 1px solid green; padding: 2px;"><math>5/3</math></span> | 0          | $14/3$      | 1          | 0          | $1/3$      | $31/3$ | $\frac{31/3}{5/3} = \frac{31}{5}$ ←      |
| 0           | $s_2$          | $-22/3$                                                                      | 0          | $32/3$      | 0          | 1          | $4/3$      | $76/3$ | $\frac{(76/3)}{(22/3)} = -\frac{76}{22}$ |
| 3           | $x_2$          | $-4/3$                                                                       | 1          | $8/3$       | 0          | 0          | $1/3$      | $10/3$ | $\frac{10/3}{-4/3} = -\frac{5}{2}$       |
| $Z_j$       |                | -4                                                                           | 3          | 8           | 0          | 0          | 1          | 10     |                                          |
| $C_j - Z_j$ |                | 3                                                                            | 0          | -11         | 0          | 0          | -1         |        |                                          |

As  $C_j - Z_j$  contains +ve value,  
we have to continue the procedure.

$\frac{31}{5}$  is the  
MINIMUM  
POSITIVE RATIO  
∴ outgoing variable is  
 $s_1$

In next table, we drop  $s_1$  &

introduce  $x_1$  as incoming variable ~~in the column~~

with its associated value  $-1$  under  $C_B$  column

from the table, observe that  $\frac{5}{3}$  is the **KEY ELEMENT**

convert the key element to unity using

the operation  $R_1 \rightarrow \frac{R_1}{(\frac{5}{3})}$

| $C_B$ | $C_j$<br>BASIS | $-1$<br>$x_1$ | $3$<br>$x_2$ | $-3$<br>$x_3$ | $0$<br>$s_1$ | $0$<br>$s_2$ | $0$<br>$s_3$ | $b$    | $\theta$ |
|-------|----------------|---------------|--------------|---------------|--------------|--------------|--------------|--------|----------|
| $-1$  | $x_1$          | $1$           | $0$          | $14/5$        | $3/5$        | $0$          | $1/5$        | $31/5$ |          |
| $0$   | $s_2$          |               |              |               |              |              |              |        |          |
| $3$   | $x_2$          |               |              |               |              |              |              |        |          |
|       |                |               |              |               |              |              |              |        |          |
|       |                |               |              |               |              |              |              |        |          |



NOW, convert the elements of **key column** to zero using the elementary row operations

$$R_2 \rightarrow R_2 + \frac{92}{3} \text{ new } R_1 \quad R_3 \rightarrow R_3 + \frac{4}{3} \text{ new } R_1$$

Also construct  $Z_j = \sum C_B a_{ij}$  &  $C_j = c_j - Z_j$  row

| $C_B$             | $C_j$<br>BASIS | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $b$   | $\theta$ |
|-------------------|----------------|-------|-------|-------|-------|-------|-------|-------|----------|
| -1                | $x_1$          | 1     | 0     | 14/5  | 3/5   | 0     | 1/5   | 31/5  |          |
| 0                 | $s_2$          | 0     | 0     | 156/5 | 22/5  | 1     | 14/5  | 354/5 |          |
| 3                 | $x_2$          | 0     | 1     | 32/5  | 4/5   | 0     | 3/5   | 58/5  |          |
| $Z_j$             |                | -1    | 3     | 82/5  | 9/5   | 0     | 8/5   | 143/5 |          |
| $C_j = c_j - Z_j$ |                | 0     | 0     | -92/5 | -9/5  | 0     | -8/5  |       |          |

clearly  $C_j - Z_j \leq 0$   
for all values



Observe the last row.

As each  $C_j - Z_j$  value is  $\leq 0$ , we can stop the procedure.

From the **Basis** column & **b** column, we have

| Basis | b       |
|-------|---------|
| $x_1$ | $31/5$  |
| $s_2$ | $354/5$ |
| $x_2$ | $58/5$  |
| $Z_j$ | $143/5$ |

∴ Optimal solution is

$$x_1 = 31/5$$

$$x_2 = 58/5$$

$$x_3 = 0 \text{ (NON BASIC)}$$

$$\& Z_{\max} = 143/5$$

$$\begin{aligned} \therefore Z_{\min} &= -Z_{\max} \\ &= -143/5 \end{aligned}$$